

Adrian Shestakov

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SUMMARY

Physicist with experience in astronomical instrumentation, scientific computing, and data analysis through work on the Vera Rubin Observatory. Developed Python-based data pipelines and analysis tools for observational astronomy research, with projects spanning gravitational microlensing, instrument characterization, and telescope commissioning.

WORK EXPERIENCE

University of California, Santa Cruz

August 2019 - January 2026

Graduate Student Researcher

Work spanned instrumentation, telescope commissioning, and survey performance modeling. Built Python-based analysis tools for CCD camera characterization, commissioning analysis, and observational data processing.

Stanford Linear Accelerator National Laboratory

October 2018 - June 2019

Physicist Intern

Supported the integration and testing of the Vera Rubin Observatory camera. Assisted with collection, documentation, and analysis of camera test data, and participated in operation of mechanical, electrical, and optical test systems during camera integration.

PROJECTS

Cosmic Superstring Microlensing in Optical Survey Data and LSST Projections [Github](#)

Developed an algorithm to search for a hypothetical transient signal in astronomical photometric time-series data. Implemented performance optimizations using profiling tools and Numba to improve computational efficiency.

Built a lightweight data processing pipeline to modularize analysis steps and support reproducible workflows across datasets. Applied the method to optical survey data and simulated LSST observations to evaluate expected event rates and survey sensitivity under a null-detection scenario.

EDUCATION

2020 - 2026 **Ph.D.** (Physics) at **University of California, Santa Cruz**

2019 - 2020 **M.Sc.** (Physics) at **University of California, Santa Cruz**

2015 - 2018 **B.Sc.** (Physics) at **University of California, Santa Cruz**

2015 - 2018 **B.A.** (Mathematics) at **University of California, Santa Cruz**

PUBLICATIONS

Snyder, Adam et al. (2020). "Laboratory measurements of instrumental signatures of the LSST camera focal plane". In: *X-Ray, Optical, and Infrared Detectors for Astronomy IX*. Ed. by Andrew D. Holland and James Beletic. Vol. 11454. International Society for Optics and Photonics. SPIE, p. 1145439. DOI: [10.1117/12.2562915](https://doi.org/10.1117/12.2562915). URL: <https://doi.org/10.1117/12.2562915>.

Quint, Bruno C. et al. (2024). "Rubin M1M3 support system dynamic performance". In: *Ground-based and Airborne Telescopes X*. Ed. by Heather K. Marshall et al. Vol. 13094. International Society for

Optics and Photonics. SPIE, p. 1309429. DOI: [10.1117/12.3019268](https://doi.org/10.1117/12.3019268). URL: <https://doi.org/10.1117/12.3019268>.

Roodman, A. et al. (2024). “LSST camera verification testing and characterization”. In: *Ground-based and Airborne Instrumentation for Astronomy X*. Ed. by Julia J. Bryant et al. Vol. 13096. International Society for Optics and Photonics. SPIE, 130961S. DOI: [10.1117/12.3019698](https://doi.org/10.1117/12.3019698). URL: <https://doi.org/10.1117/12.3019698>.

Stalder, Brian et al. (2024). “Rubin Observatory Simonyi Survey Telescope integrated mount performance”. In: *Ground-based and Airborne Telescopes X*. Ed. by Heather K. Marshall et al. Vol. 13094. International Society for Optics and Photonics. SPIE, p. 1309409. DOI: [10.1117/12.3019266](https://doi.org/10.1117/12.3019266). URL: <https://doi.org/10.1117/12.3019266>.

SKILLS

Programming Languages and Platforms: Python, SQL, Git, GitHub, Jupyter

Python Libraries: Numpy, Scipy, Matplotlib, Seaborn, Pandas, Numba

Instrumentation: CCD Characterization, Telescope Commissioning

Scientific Computing: Data Processing Pipelines, Statistical Data Analysis, Time-Series Analysis, Numerical Modeling, Algorithmic Optimization, Monte Carlo Simulation